Sharath S Hebbar

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Skills

Frameworks and Libraries: PyTorch, TensorFlow, Keras, Flask, FastAPI, Pandas, HuggingFace.
Languages and Databases: Python, C/C++, SQL, Firestore/Firebase.
ML Techniques and Tools: Supervised, Unsupervised, Neural Networks, NLP, Deep Learning, Transfer Learning, Prompt Engineering, Generative AI, Agentic AI, RAG, Langchain, CrewAI, LangGraph, FineTuning Language Models
Platform and DevOps: Git, GitHub, GitLab, Google Cloud Platform, VertexAI, DocumentAI, Azure Cloud, Databricks, Docker.
Soft Skills: Client Handling, Call Planning, Communicating to broader audience.

Experience

Data Scientist Jan 2024 – Present
IQVIA *Bengaluru, Karnataka, India*

- Integrated **GPT-4o** into **PowerBI** using a **Agentic AI framework** to enable advanced **summarization** and interactive **conversations** with data visualizations, significantly enhancing data interpretation and accelerating decision-making processes.
- Built an end-to-end **Conversational Intelligence Pipeline** to process Med Rep to HCP audio interactions, with modules for transcription, translation, medical correction, speaker diarization, and summarization. Leveraged **LLMs** for domain-specific correction of **medical terms**, **role-based speaker assignment**, and summarization to extract **Agreed Actions, Areas of Interest, and Objections**.
- Conducted comprehensive **Exploratory Data Analysis (EDA)** on large-scale pharma data of more than **100K+ records** (one region) to extract valuable insights about various promotional channels and HCPs' engagement.
- Calculated **HCP Potential using 8–10 engagement channels, prescriptions, and sales data**, and applied **clustering** to segment HCPs by behavior and interaction patterns. Developed targeted strategies that improved campaign focus, achieving **70–80% accuracy** in identifying the right **Healthcare Professionals** for outreach.
- Collaborated with the Data Engineering team to **optimize, industrialize, and scale code solutions** across **12+ regions**, integrating them into data pipelines to ensure **scalability, production efficiency**.

Senior Machine Learning Engineer Aug 2022 – Jan 2024
Quantiphi Analytics *Bengaluru, Karnataka, India*

- **Mentored and Spearheaded** a team of **7 ML Interns** to build a Proof of Concept based on a **Multi-modal search engine**.
- Developed a **customer attrition** model with **80% F-score** using a large dataset of 2 million records with Machine learning algorithms such as **XGBoost, Random Forest, CatBoost, and Artificial Neural Networks** to predict **customer churn**.
- Built a **Custom Synthetic Document Generator** which can be used to generate synthetic documents for training ML models.
- Trained a custom processor in **Google Cloud Document AI** using synthetic data generated by the **Custom Synthetic Document Generator (CSDG)**, automating document tagging and **reducing annotation time and manual effort by 50%**.
- Worked on a proof-of-concept to implement **Sentiment Analysis** using state-of-the-art Transformer models, including **BERT, RoBERTa, DistilBERT, and XLNET** achieving a performance of **80%**.

Projects

Automated Custom Segments for Enhanced Targeting: Microsegmentation | *SQL, PySpark, Python, Clustering, GenAI, LLM, Big data*

- Designed and implemented a data-driven segmentation solution using **SQL, Python, PySpark, and K-Means Clustering** to categorize Healthcare Professionals (HCPs) based on their engagement and activity patterns.
- Created a customer engagement score to evaluate HCPs' engagement, utilized **prescription, and sales potential** to calculate HCPs potential. Segmented HCPs using **K-Means Clustering** based on Engagement, Prescription, and Sales potential.
- Deployed the system across multiple markets, enhancing targeted marketing strategies and improving HCP engagement, resulting in more efficient targeting and better outcomes.

Power AI | *Python, GenAI, Prompt Engineering, PowerBI, Langchain, Agentic Framework, CrewAI, LangGraph, Text-to-SQL*

- Integrated **AI into PowerBI** using **Python, GPT-4, and a Agentic Framework** to enable advanced summarization and interactive data conversations.
- Developed a **Agentic AI framework** that ingests structured data from **PowerBI** sources, generates relevant questions, dynamically creates and executes code to retrieve answers, and compiles **Q&A pairs** to automate chart-level and dashboard summaries, enhancing rapid insight extraction and improving data comprehension by over **60%**.
- Built a custom chatbot to deliver **tailored, actionable business insights**, significantly improving user experience, decision-making speed, and dashboard accessibility.

Education

University Visvesvaraya College of Engineering Bengaluru, Karnataka, India
*Bachelor of Technology in Computer Science (**9.45 CGPA**)* *Sep 2019 – May 2022*

MEI Polytechnic Bengaluru, Karnataka, India
*Diploma in Computer Science and Engineering (**90.03%**)* *July 2016 – June 2019*